Topographic Projection Simulation: Requirements Specification Document

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1 Foreword

This simulation program is designed to advance the understanding of axon guidance and topographic mapping, which are critical processes in neural development. Although the primary focus is on retinotectal projections, the program is adaptable to other areas, providing a versatile tool for the scientific community to explore and experiment.

This document outlines the functional and non-functional requirements, laying the foundation for effective use and ongoing development.

2 Target Definition

2.1 Scope of application

The primary goal of this simulation program is to serve as a research tool for axon guidance studies, particularly for experimenting with models developed for topographic mapping in retinotectal simulations and other contexts. Equipped with customizable parameters the program enables researchers to investigate the dynamics of eph/ephrin signal molecules. Researchers can adjust variables, such as growth cone size, step distance, and substrate types, while visualizing outcomes, including growth cone trajectories and projection mappings.

The product is a ready-to-use tool that can be easily extended or used as a library for other projects such as web applications offering a graphical user interface (GUI).

2.2 Target audience

The target audience includes individuals who develop or use computational simulations of neurodevelopmental models, primarily scientists and researchers.

To use this product, users should have a basic understanding of Git for cloning the repository and Python for building the project, configuring parameter values, and running the program.

3 Functional Requirements

/FR-10/ Simulation

The program conducts simulations of the Topographic Projection [1] based on user-defined configurations. The simulation is divided into several key steps:

1. Initialization
2. Iteration
    a) Step Proposal Generation
    b) Potential Calculation
    c) Step Decision Process
    d) Adaptation Mechanism
3. Completion

/FR-11/ Initialization

The simulation initializes by setting up all growth cone objects and the substrate (see 3), along with simulation logic configuration (See 3 and 3). Growth cones are placed evenly spaced at the anterior edge of the substrate.

/FR-12/ Step Proposal Generation

For each growth cone, the program iterates over a specified number of steps to generate random step proposals. These proposals are generated based on predefined probabilities for movement in the x and y directions. Each step proposal considers the substrate's boundaries to ensure valid movement.

/FR-13/ Potential Calculation

For each step proposal, the simulation calculates both the current guidance potential and the hypothetical potential at the proposed new location. This involves using the growth cone's position to assess potential interactions with the substrate and other growth cones.

/FR-14/ Adaptation Mechanism

If adaptation is enabled, the simulation applies an adaptation mechanism to each growth cone, dynamically adjusting its sensitivity based on past interactions and predefined adaptation parameters.

/FR-15/ Step Decision Process

The program decides whether to accept each step proposal by comparing the guidance potentials of the current and proposed positions. This decision is probabilistic, based on the calculated probabilities of potential changes. If the force parameter is enabled, steps are accepted unconditionally.

/FR-16/ Completion

The simulation completes after iterating through all the specified steps for each growth cone. The final state of the growth cones is recorded and returned as the simulation result.

/FR-20/ Simple Simulation Parameters

Users can customize the following basic simulation parameters:

1. Number of growth cones (`GC_COUNT`): Set the number of growth cones in the simulation. Must be positive integer.
2. Size of growth cones (`GC_SIZE`): Define the size of each growth cone. Must be positive integer.
3. Total number of steps (`STEP_NUM`): Specify the total number of steps for the simulation. Must be positive integer.
4. Distance covered in each step (`STEP_SIZE`): Determine the distance covered in each simulation step. Must be positive integer.

/FR-30/ Advanced Simulation Parameters

Users can configure the following advanced simulation parameters:

1. X and Y step probabilities (`X_STEP_POSSIBILITY`, `Y_STEP_POSSIBILITY`): Set the probabilities of steps taken in the x and y directions. Must be double between 0 and 1.
2. Sigma for Gaussian distribution (`SIGMA`): Controls the randomness in step decisions. Must be non-negative double.
3. Force along X axis (`FORCE`): Enable or disable force along the x-axis. Must be boolean.
4. Adaptation mechanism: Configure the adaptation mechanism to adjust growth cone behavior dynamically. Values detailed in FR-32.

5. Sigmoid function: Adjust sigmoid function parameters to model non-linear fiber-fiber interaction strength. Values detailed in FR-31.
6. Interaction switches: Enable or disable various interaction types to simulate complex neuronal interactions. Values detailed in FR-33.

/FR-31/ Sigmoid Function Parameters

Users can adjust the following sigmoid function parameters:

1. Sigmoid steepness (`SIGMOID_STEEPNESS`): Control the steepness of the sigmoid function. Must be double.
2. Sigmoid shift (`SIGMOID_SHIFT`): Adjust the shift of the sigmoid curve. Must be double.
3. Sigmoid height (`SIGMOID_HEIGHT`): Set the maximum height of the sigmoid function. Must be double.

/FR-32/ Adaptation Parameters

Users can configure the following adaptation parameters:

1. Adaptation enabled (`ADAPTATION_ENABLED`): Enable or disable adaptation in the simulation. Must be boolean.
2. Mu for adaptation (`ADAPTATION_MU`): Set the coefficient (strength) for adaptation. Must be non-negative double.
3. Lambda for adaptation (`ADAPTATION_LAMBDA`): Set the resetting force for adaptation. Must be non-negative double.
4. Adaptation history length (`ADAPTATION_HISTORY`): Set the length of adaptation history. Must be non-negative integer.

/FR-33/ Interaction Switches

Users can enable or disable the following interaction switches:

1. Fiber-fiber interaction (`FF_INTER`): Enable or disable interactions between fibers. Must be boolean.
2. Fiber-target interaction (`FT_INTER`): Enable or disable interactions between fibers and targets. Must be boolean.
3. Forward signal (`FORWARD_SIG`): Enable or disable forward signaling. Must be boolean.
4. Reverse signal (`REVERSE_SIG`): Enable or disable reverse signaling. Must be boolean.

/FR-40/ Substrate Parameters

Users can customize the following substrate parameters:

1. Dimensions of the substrate (`ROWS`, `COLS`): Set the number of rows and columns for the substrate. Must be positive integer.
2. Type of substrate (`SUBSTRATE_TYPE`): Choose the type of substrate for the simulation. Must be string matching one of the types in FR-41.
3. Further type specific parameters. Detailed from FR-42 to FR-45.

/FR-41/ Supported Substrate Types

The following substrate types are supported by default:

1. Continuous gradients
2. Wedges
3. Stripe assay
4. Gap assay

/FR-42/ Continuous Gradient Parameters

Users can customize the following continuous gradient parameters:

1. Continuous signal start (`CONTINUOUS_SIGNAL_START`): Set the starting value of the continuous gradient signal. Must be double.
2. Continuous signal end (`CONTINUOUS_SIGNAL_END`): Set the ending value of the continuous gradient signal. Must be double.

/FR-43/ Wedges Parameters

Users can customize the following wedges parameters:

1. Wedge narrow edge (`WEDGE_NARROW_EDGE`): Define the narrow edge width of wedges. Must be positivie integer.
2. Wedge wide edge (`WEDGE_WIDE_EDGE`): Define the wide edge width of wedges. Must be positive integer.

/FR-44/ Stripe Assay Parameters

Users can customize the following stripe assay parameters:

1. Stripe forward (`STRIPE_FWD`): Enable or disable the forward stripe assay. Must be boolean.
2. Stripe reverse (`STRIPE_REW`): Enable or disable the reverse stripe assay. Must be boolean.

3. Stripe concentration (`STRIPE_CONC`): Set the concentration of stripes. Must be double.
4. Stripe width (`STRIPE_WIDTH`): Define the width of stripes. Must be positive integer.

/FR-45/ Gap Assay Parameters

Users can customize the following gap assay parameters:

1. Gap begin (`GAP_BEGIN`): Set the starting position of the gap. Must be double between 0 and 1.
2. Gap end (`GAP_END`): Set the ending position of the gap. Must be double between 0 and 1 and bigger than begin.
3. Gap first block (`GAP_FIRST_BLOCK`): Specify the type of the first block in the gap assay. Must be string matching either LIGAND or RECEPTOR.
4. Gap second block (`GAP_SECOND_BLOCK`): Specify the type of the second block in the gap assay. Must be string matching either LIGAND or RECEPTOR.

/FR-50/ Visualization

The program visualizes the following:

1. Substrate
2. Tectum end-points of growth cones on empty background or substrate
3. Projection Mapping on a two-dimensional plane, with axes representing the initial and final growth cone positions
4. Growth cone initial sensor values *(new)*
5. Growth cone trajectory
6. Growth cone history *(new)*

/FR-51/ Evaluation of Projection

The program calculates linear regression on projection mapping and presents its graphical representation, including correlation, intercept, and slope values.

/FR-52/ Evaluation of Projection: Smooth Interpolation

User can select smooth interpolation instead of linear regression. *(new)*

/FR-60/ Result Saving (optional)

Users can save simulation results textually as txt file containing all values of each Growth Cone object or visually as png file as specified in FR-50.

/FR-70/ Estimated Time

During the simulation, the program displays a progress indicator based on the ratio of the current step to the total steps. After the first 1,000 steps, the program will estimate the remaining time based on the duration of these initial steps and updates this estimate after each subsequent 1,000 steps.

4 Non-Functional Requirements

/NR-10/ Scalability

The program should be designed to accommodate up to 1,000 growth cones, support a total number iteration steps up to 10^6, and handle field sizes consisting of thousands of columns and rows.

/NR-20/ Performance

Considering NR-10, the program should optimize simulation time to efficiently handle larger inputs by employing parallel computing algorithms and implementing early stopping mechanisms.

/NR-30/ Usability

The program should be intuitively usable by bioscientists with minimal programming knowledge.

/NR-40/ Extensibility

The program should be designed for easy extensibility, allowing for the addition of new substrate types, novel configuration methods for substrates and growth cones, new visualization methods and the implementation of new experiments. It should provide clear entry and exit points, enabling integration with other applications, such as those offering graphical user interfaces (GUI) or web applications, to use it as a library.

5 Constraints

/CN-10/ Development Platform

The development platform is limited to Python [2].

/CN-20/ Mathematical Model

All calculations in the simulation will adhere to predetermined mathematical models of this Paper [3].

6 Product Environment

6.1 Software Environment

/SE-10/ Operating System

The program is compatible with Windows, macOS, and Linux operating systems.

/SE-20/ Programming Language

The simulation is developed in Python. Users should have Python version 3.8 or higher installed.

/SE-30/ Dependencies

The program relies on several Python libraries, including:

- `NumPy`: for numerical computations.
- `Matplotlib`: for data visualization and plotting.
- `SciPy`: for scientific computing and technical functionality.

These dependencies can be installed using package managers such as `pip`.

/SE-40/ Version Control

Git is used for version control. Users should have Git installed to clone the repository and manage version control efficiently if they would like to extend the program.

6.2 Hardware Environment

/HE-10/ Processor

A multi-core processor like `Intel(R) Core(TM) i5-12400F` is recommended for efficient computation, especially when simulating a large number of growth cones.

/HE-20/ Memory

At least 4 GB of RAM is recommended to handle simulations with large inputs efficiently.

/HE-30/ Storage

A minimum of 500 MB of free disk space is required for installing the program and its dependencies. Additional space may be needed for storing simulation data and results.

7 Case Study: Continuous Gradients

This section demonstrates the application of the simulation program on a continuous gradient substrate. The results illustrate how growth cones behave in a controlled environment, highlighting the program's capabilities in modeling complex biological systems.

7.1 Simulation Setup

The simulation was conducted with the following parameters:

- `GC_COUNT`: 25
- `GC_SIZE`: 3
- `STEP_SIZE`: 1
- `STEP_NUM`: 8000
- `X_STEP_POSSIBILITY`: 0.55
- `Y_STEP_POSSIBILITY`: 0.50
- `SIGMOID_STEEPNESS`: 4
- `SIGMOID_SHIFT`: 3
- `SIGMA`: 0.12
- `FORCE`: False
- `FORWARD_SIG`: True
- `REVERSE_SIG`: True
- `FF_INTER`: True
- `FT_INTER`: True
- `ADAPTATION_ENABLED`: True
- `ADAPTATION_MU`: 0.006
- `ADAPTATION_LAMBDA`: 0.0045
- `ADAPTATION_HISTORY`: 50
- `SUBSTRATE_TYPE`: Continuous Gradients
- `ROWS`: 100
- `COLS`: 100
- `CONTINUOUS_SIGNAL_START`: 0.01
- `CONTINUOUS_SIGNAL_END`: 1

7.2 Results

The simulation results demonstrate that growth cones respond consistently to continuous gradients by migrating toward regions with lower guidance potential. This behavior indicates that the growth cones are locating substrate areas that match their molecular signatures. The following figures illustrate the initial states of growth cones and substrates, as well as their final positions, mappings, and movement trajectories.

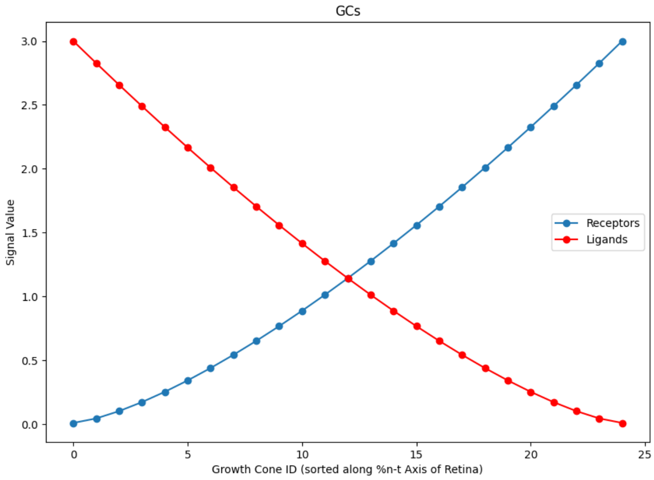

Figure 1: Growth Cone Initial Signal Values

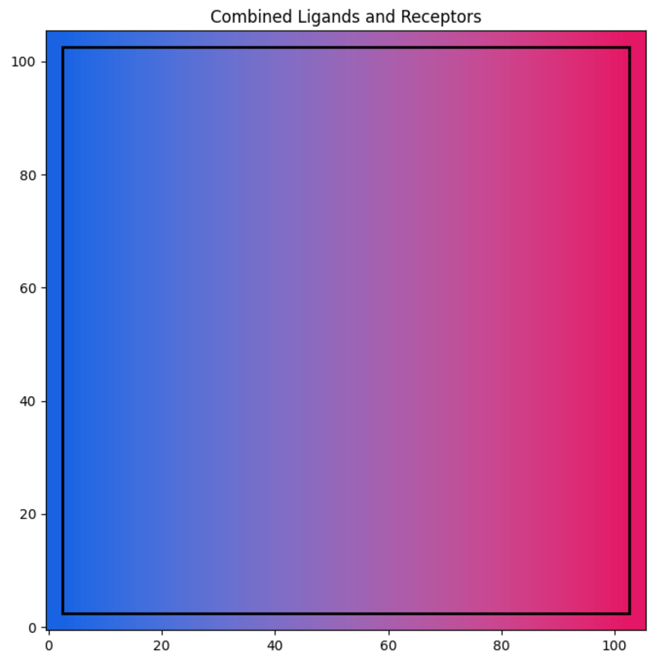

Figure 2: Combined Ligands and Receptors on Substrate

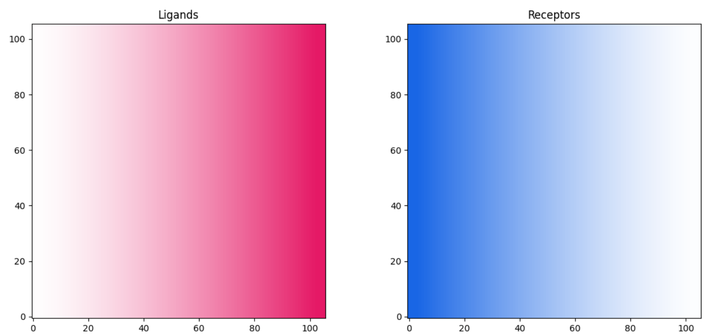

Figure 3: Separated Visualization of Ligands and Receptors

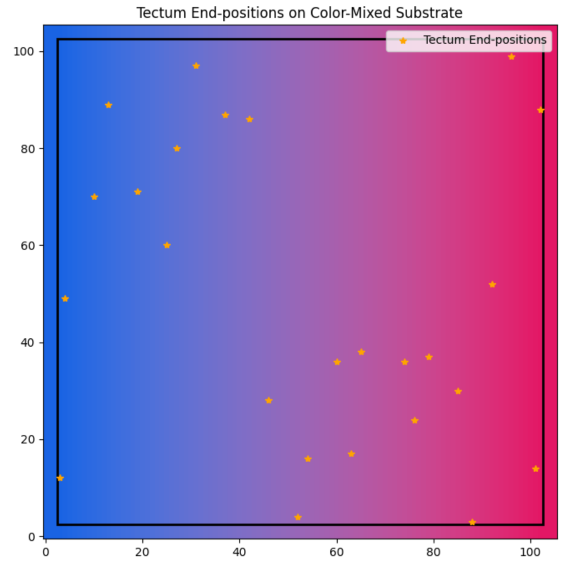

Figure 4: Tectum End-positions on Color-Mixed Substrate

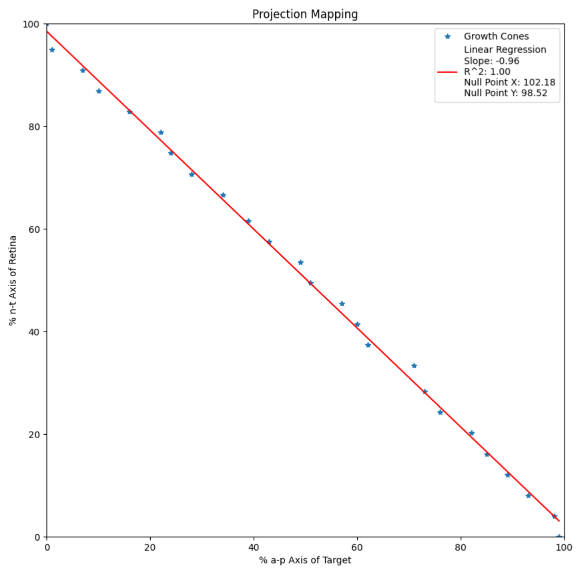

Figure 5: Projection Mapping with Linear Regression

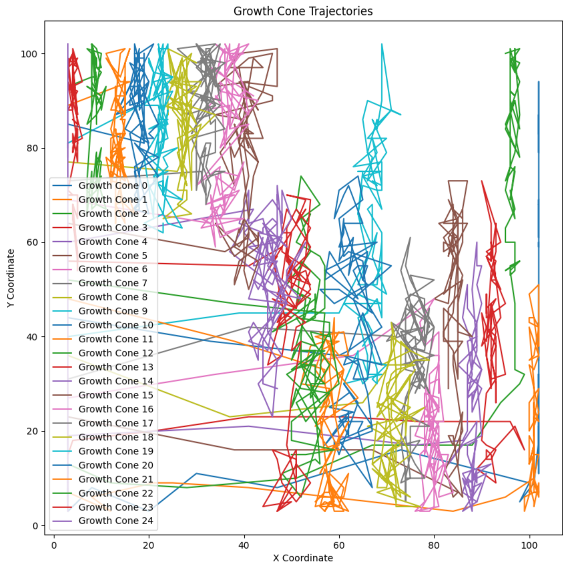

Figure 6: Growth Cone Trajectories

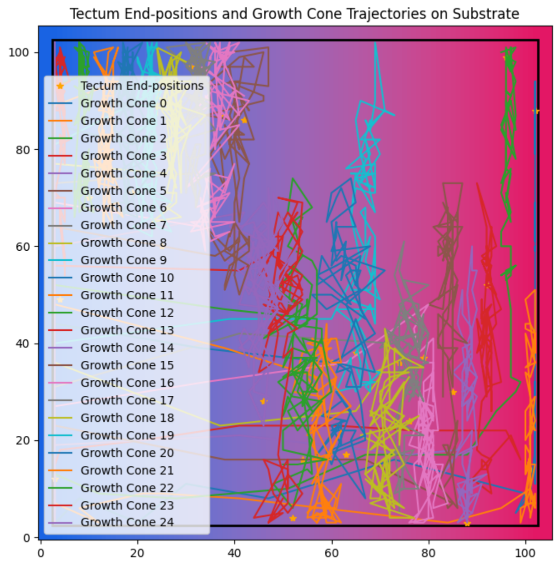

Figure 7: Tectum End-positions and Growth Cone Trajectories on Substrate

8 Glossary

Axon Guidance: The process by which neurons send out axons to reach the correct targets, critical for neural circuit formation.

Topographic Mapping: The ordered projection of a sensory surface (e.g., retina) onto a neural target (e.g., tectum) such that spatial relationships are preserved.

Retinotectal Projections: The neural pathways from the retina to the tectum in the brain, used as a model system for studying neural development and axon guidance.

Growth Cone: A dynamic, motile structure at the tip of a growing axon or dendrite, which explores the environment to find its path.

Substrate: The physical or chemical surface on which growth cones move during neural development.

Sigmoid Function: A mathematical function having an "S" shaped curve (sigmoid curve) often used to model growth and neural response functions.

Gaussian Distribution: A probability distribution used to model randomness, characterized by a symmetric bell-shaped curve centered around a mean value.

Adaptation Mechanism: A process in the model that allows growth cones to change their behavior based on past experiences and current conditions.

Fiber-Fiber Interaction: The interactions between axonal fibers that influence their growth and final connectivity.

Fiber-Target Interaction: The interactions between axonal fibers and their target cells, affecting growth direction and connectivity based on target-derived signals.

Forward Signal: The signaling process where Eph receptors on axonal growth cones interact with ephrin ligands in the environment, leading to repulsive cues that help guide the axon away from areas with high ephrin concentration.

Reverse Signal: The signaling process where ephrin ligands on growth cones receive signals from Eph receptors in the environment, leading to repulsive cues that help guide the axon away from areas with high Eph concentration.

Continuous Gradients: Gradual changes in chemical concentration across the substrate that guide axon growth direction.

Stripe Assay: An experimental setup used to study axon guidance, where alternating stripes of different substrates or signals are presented to growing axons.

Wedges: A type of substrate pattern used to study axon guidance, where the substrate varies in width across the surface.

Gap Assay: An experimental design where a gap is introduced in the substrate to study the response of growing axons to interruptions in their path.

Tectum: A part of the brain involved in visual processing, often studied in the context of retinotectal mapping.

Projection Mapping: The visualization of how neural projections map from one area, such as the retina, to another, such as the tectum.

Nasal (Retinal Region): Refers to the area of the retina closer to the nose. In retinotectal mapping, nasal retinal axons typically project to the anterior part of the tectum.

Temporal (Retinal Region): Refers to the area of the retina closer to the temples (sides of the head). Temporal retinal axons generally project to the posterior part of the tectum in retinotectal mapping.

Anterior (Tectal Region): The front part of the tectum. In retinotectal projections, anterior corresponds to the region that receives inputs from nasal retinal axons.

Posterior (Tectal Region): The back part of the tectum. In retinotectal mapping, posterior is the region that receives inputs from temporal retinal axons.

References

[1] Gebhardt C, Bastmeyer M, Weth F. Balancing of ephrin/Eph forward and reverse signaling as the driving force of adaptive topographic mapping. Development. 2012 Jan;139(2):335-45. doi: 10.1242/dev.070474. Epub 2011 Dec 7. PMID: 22159582.

[2] https://www.python.org/

[3] Felix Fiederling, Markus Weschenfelder, Martin Fritz, Anne von Philipsborn, Martin Bastmeyer, Franco Weth (2017) Ephrin-A/EphA specific co-adaptation as a novel mechanism in topographic axon guidance eLife 6:e25533